

20 B $R = \rho L / A , R = V / I$

$$V/I = \rho L / A$$

$$L = VA / I\rho$$

$$= 240 [\pi(3.0 \times 10^{-4})^2] / (2.0 \times 1.4 \times 10^{-6})$$

$$= 24 \text{ m}$$